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Financial Performance and Social Responsibility in Mission-Driven Firms

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1 Introduction

To understand the emergence and consolidation of Corporate Social Responsibility (CSR), it is essential to consider the historical and economic transformations in the Western world after World War II. The post-war period saw rapid industrial expansion, the growth of multinational corporations, and significant socio-political changes, including environmental awareness, civil rights movements, and anti-war protests. These developments intensified public scrutiny of corporate power and laid the foundations for CSR, as businesses were expected to assume responsibilities beyond profit maximization. Regulatory measures in the 1970s, such as environmental and labour protection laws, reinforced this notion. Carroll's (1979) multidimensional framework—economic, legal, ethical, and discretionary responsibilities—consolidated CSR conceptually. From the 1980s onward, globalization and market liberalization redefined corporate accountability, with multinational firms facing reputational pressures and diverse stakeholder expectations, promoting CSR's institutionalization and strategic integration. By the 2000s and 2010s, CSR evolved toward a value-creation perspective, particularly through the concept of shared value, linking economic competitiveness with social progress (Porter & Kramer, 2006; 2011). This evolution has also taken concrete legal forms: in France, the **mission-driven company** (*société à mission*) status, introduced by the PACTE Law of 2019, allows a company to enshrine a specific purpose and social or environmental objectives in its by-laws, the achievement of which is verified by an independent third-party body[1]. Overall, CSR has shifted from philanthropy to regulatory compliance and strategic integration, reflecting changing societal expectations and the recognition that long-term economic performance is intertwined with social and environmental responsibility (Agudelo et al., 2019).

Today, CSR has become increasingly comprehensive and institutionalized, supported by internationally recognized standards and global initiatives such as the United Nations Global Compact[2]. To clearly define this concept, we rely on the principles promoted by the United Nations Global Compact on its website. It is based on internationally recognized frameworks such as ISO 26000, the United Nations Guiding Principles on Business and Human Rights, and the OECD Guidelines for Multinational Enterprises. According to the European Commission, CSR means that companies must take re-

[1]Annuaire des Entreprises (2024). *Société à mission*. Retrieved from https://annuaire-entreprises.data.gouv.fr/definitions/societe-a-mission

[2]United Nations Global Compact – France Network (2024). *Qu'est-ce que la RSE ?*.

sponsibility for the effects of their activities on society. This requires compliance with existing legislation and collective agreements, but it also goes beyond legal obligations. CSR involves integrating social, environmental, ethical, human rights, and consumer concerns into business operations and core strategy, in close collaboration with stakeholders.

Being socially responsible implies respecting applicable laws and regulations, identifying and mitigating negative impacts of business activities, investing in human capital and environmental protection, and contributing to sustainable development objectives.

CSR is closely aligned with the United Nations Sustainable Development Goals (SDGs). It serves as a framework that encourages companies to rethink management and production models, fostering economic performance while also advancing social equity and environmental sustainability. In this sense, CSR and the SDGs mutually reinforce one another, positioning businesses as key actors in addressing global challenges such as inequality, poverty, and climate change.

Building on this understanding, this study investigates whether mission-driven company certification has a measurable impact on firms' financial performance over the 2019–2024 period. To address this question, we first review the existing literature to identify key theoretical arguments and empirical findings, and to formulate research hypotheses. An empirical analysis is then conducted using a dataset of firms, applying appropriate econometric methods to test the relationships between CSR engagement and financial performance. The study concludes by comparing the empirical results with the expectations derived from prior research.

2 Literature review

Corporate Social Responsibility and Financial Performance

The relationship between Corporate Social Responsibility (CSR) and corporate financial performance (CFP) has been extensively debated in management, finance, and economics literature. While early theoretical discussions framed CSR as either a constraint on profitability or a potential source of competitive advantage, subsequent empirical studies have produced mixed and sometimes contradictory results. This section synthesizes the main theoretical foundations and empirical findings to clarify the mechanisms through which CSR may influence financial performance.

2.1 Theoretical Foundations

The modern debate on CSR originates from two opposing theoretical perspectives.

2.1.1 Shareholder Primacy

In a classical view, articulated by Friedman (1970), the sole responsibility of business is to increase profits within the bounds of the law. From this perspective, CSR expenditures represent a misallocation of corporate resources unless they directly contribute to shareholder value. CSR may therefore reduce financial performance if it imposes additional costs without corresponding revenue gains.

2.1.2 Stakeholder Theory

In contrast, Freeman (1984) developed stakeholder theory, arguing that firms must create value for a broad set of stakeholders, not only shareholders. Under this framework, CSR strengthens relationships with employees, customers, suppliers, regulators, and communities, thereby enhancing long-term firm performance.

Building on this logic, Carroll (1979) proposed a multidimensional model of corporate responsibility encompassing economic, legal, ethical, and discretionary responsibilities. This framework laid the conceptual foundation for linking social and financial performance.

The theoretical tension between shareholder primacy and stakeholder theory continues to structure the CSR–CFP debate.

2.2 Early Empirical Evidence and the Problem of Inconsistency

Empirical research began intensifying in the 1990s, but results quickly proved inconclusive.

Waddock and Graves (1997) found a positive relationship between corporate social performance and financial performance, suggesting that socially responsible firms may outperform others financially.

However, McWilliams and Siegel (2000) argued that previous findings suffered from model misspecification. After controlling for R&D intensity and other omitted variables, the relationship between CSR and financial performance became statistically insignificant. Their work introduced methodological caution into the literature.

To clarify conflicting findings, Margolis and Walsh (2003) conducted a large-scale review and concluded that while many studies report a positive association, the overall effect is modest.

A major contribution came from Orlitzky et al. (2003) whose meta-analysis found a generally positive relationship between CSR and financial performance. Importantly, they highlighted that the relationship is stronger when CSR is measured through reputation-based indicators rather than purely objective measures.

Beyond correlation, the question of causality remains central. Waddock and Graves (1997) postulate the existence of a virtuous circle: if CSR improves financial performance (good management theory), solid financial performance also frees up surplus resources that can be invested in CSR (slack resources theory). This dual causality, or endogeneity, suggests that CSR and financial performance reinforce each other in the long term.

These studies collectively demonstrate that although a positive association is frequently observed, methodological choices significantly influence results.

2.3 CSR as a Strategic Resource

Later research shifted from asking whether CSR affects performance to understanding how it does so.

Eccles et al. (2014) showed that firms adopting high sustainability practices tend to outperform over the long term. They argued that CSR reshapes organizational processes, governance structures, and stakeholder engagement, thereby improving resilience and long-term profitability.

Similarly, Fatemi et al. (2018) found that ESG[3] performance enhances firm value, particularly when disclosure quality is high. Their findings emphasize the role of transparency and information in capital markets.

This stream of literature views CSR as a strategic asset capable of generating reputational capital, reducing information asymmetry, and improving investor confidence.

2.4 Governance, Cost of Capital, and Moderating Mechanisms

Another important line of research examines the interaction between CSR and corporate governance.

Jo and Harjoto (2011) show that CSR engagement is positively associated with firm value and that governance quality influences CSR decisions.

In addition, El Ghoul et al. (2011) demonstrate that firms with strong CSR performance benefit from a lower cost of equity capital. This suggests that financial markets reward socially responsible behavior, possibly due to lower perceived risk.

More recently, the focus has shifted to the reliability of non-financial information. With the entry into force of strict regulations such as the CSRD in Europe, the financial market is becoming more sensitive to the risk of greenwashing. Huang et al. (2022) suggest that the positive effect of CSR only holds true if there is complete transparency, with investors now penalizing companies whose social commitments are not corroborated by auditable data.

This body of work suggests that CSR does not operate in isolation but interacts with governance structures and financial market perceptions.

[3] ESG (Environmental, Social, and Governance) criteria constitute the operational and measurable dimension of CSR, translating its broad principles into specific indicators used by investors and rating agencies to assess non-financial corporate performance. See European Commission, *Corporate Sustainability and Responsibility*, https://single-market-economy.ec.europa.eu/industry/sustainability/corporate-sustainability-and-responsibility_en.

2.5 The Multidimensional Nature of CSR

A critical refinement in the literature concerns the multidimensionality of CSR.

Cavaco and Crifo (2014) argue that CSR dimensions may be complementary or substitutable. Using panel data analysis, they find that certain CSR practices (such as human resource policies and business behavior) reinforce each other in generating financial performance, while others may involve trade-offs.

A major advance in this multidimensional perspective is the concept of financial materiality. Khan et al. (2016) demonstrate that the impact of CSR on stock market value depends on the sectoral relevance of the issues addressed. Thus, an environmental investment will have a more significant financial impact for an industrial company than for a service company, highlighting that only "material" CSR generates a measurable competitive advantage.

This perspective helps explain previous inconsistencies in empirical findings. If CSR dimensions interact, aggregating them into a single score may obscure heterogeneous effects.

2.6 Synthesis and Research Implications

The literature suggests several key conclusions:

- Theoretical perspectives differ, but stakeholder theory provides stronger support for a positive CSR–CFP relationship.
- Empirical evidence generally indicates a positive but modest association.
- Methodological rigor is crucial, particularly regarding endogeneity and measurement issues.
- CSR effects depend on governance quality, disclosure, and multidimensional interactions.
- Long-term performance measures often capture CSR benefits better than short-term indicators.

Based on stakeholder theory (Freeman, 1984) and empirical evidence suggesting a generally positive association between CSR and financial performance (Waddock & Graves, 1997; Orlitzky et al., 2003; Eccles et al., 2014), CSR can be analyzed as a strategic investment. It enhances the value of the firm by optimizing stakeholder relations, consolidating reputational capital, and reducing information asymmetry.

However, recent literature emphasizes that this link is not automatic: it depends on the sectoral materiality of the issues (Khan et al., 2016) and the quality of non-financial transparency, which has become crucial under the pressure of regulations such as the CSRD (Huang et al., 2022). Although some studies point to methodological biases or neutral effects when innovation (R&D) is controlled (McWilliams & Siegel, 2000), the mainstream view is that authentic and auditable CSR commitment reduces the cost of capital and improves long-term resilience.

Therefore, considering CSR no longer as a discretionary cost but as a lever for overall performance, we formulate the following hypothesis:

H1: High levels of Corporate Social Responsibility are positively associated with companies' financial performance.

In the following sections, we will verify this hypothesis in light of the facts.

3 Empirical study

3.1 Construction of the Dataset

The dataset used in this study was constructed from information collected through web scraping from several reliable sources. The first step consisted in selecting the companies to be included in the analysis, focusing specifically on French mission-driven companies. To do so, we relied on an initial database of **2,221 mission-driven companies** obtained from the website of the *Observatoire des Sociétés à Mission*[4].

This database contained several essential descriptive variables used to identify and characterize the companies, including: the **SIREN number**, the **legal name**, the **trade name**, the **mission-driven status**, the **date of incorporation**, the **NAF/APE code**, the **description of the activity**, and the **date on which the mission was formally adopted** by shareholders.

These elements formed the foundation of our enrichment work prior to integrating financial and CSR information.

In a second phase, we collected the financial data required to study the relationship between economic performance and CSR engagement. The choice of variables is based on the literature identifying the most representative indicators of corporate financial performance (Sahut et al., 2018; Abidi & Diaye, 2023). The financial data were automatically extracted from *Pappers*, an open-access platform aggregating financial and legal documents on French companies. The variables collected include **revenue**, **operating income**, **net income**, **equity**, **financial debt**, and **return on assets (ROA)**. We also calculated the **Return on Equity (ROE)**, defined as the ratio of net income to equity, in order to obtain a standardized and comparable measure of financial profitability.

The collection of financial data relied on an automated **ETL pipeline** designed to transform raw web data into a reliable and exploitable analytical dataset. This pipeline consists of four main stages: extraction, normalization, feature engineering, and integration.

[4] https://www.observatoiredessocietesamission.com/

1. Data extraction: Extraction was performed using a Python script based on Selenium WebDriver, which simulates human navigation on the *Pappers.fr* platform. An automatic scrolling function ensured the full loading of financial tables, which are often displayed at the bottom of the page. In addition, a batching system was implemented to divide SIREN numbers into groups and ensure process stability. In case of interruption, the pipeline could resume without losing progress. This stage produced heterogeneous raw data directly extracted from HTML.

2. Normalization and standardization: The extracted financial values appeared in various formats (K, M, Bn). A systematic normalization process was therefore applied to ensure unit consistency and avoid scale errors. The `nettoyer_valeur` function converted each value into euros according to the formula:

$$\text{Final value} = \text{Raw value} \times \text{Unit multiplier}$$

For example: 1.25 M€ = 1,250,000 €. This step ensured data homogeneity and prepared the dataset for analytical computations.

3. Feature engineering: Based on the normalized data, several financial indicators were constructed to enrich the dataset and facilitate statistical analyses. These include:

- **ROA (Return on Assets)**:

$$\text{ROA} = \left(\frac{\text{Net income}}{\text{Total assets}}\right) \times 100$$

- **Squared net assets**:

$$(\text{Equity})^2$$

  which captures potential nonlinear effects;

- **Gearing**, an indicator of financial leverage and dependence on external debt.

4. Integration and merging: The newly extracted data were merged with the initial database to avoid redundant queries and enrich existing observations. The pair (**SIREN**, **year**) served as the merging key to ensure reliable alignment between sources.A final cleaning step removed rows lacking exploitable information before generating a consolidated Excel file.

In a third phase, we constructed the CSR variables necessary to characterize the non-financial profile of each firm in the sample. Following the same logic as for financial data, an automated pipeline was developed to assign each firm a CSR component, a sub-theme, and a quantitative benchmark score derived from the EcoVadis[5] sectoral framework.

The choice of EcoVadis as the reference framework for CSR scoring is motivated by several methodological and practical considerations. First, EcoVadis constitutes one of the most widely recognized and externally audited CSR rating systems in the European context, covering over 130,000 companies across more than 200 industry categories. Its sectoral granularity — structuring assessments around four pillars (Environment, Labor & Human Rights, Ethics, and Sustainable Procurement), aligns closely with the multidimensional conception of CSR adopted in this study (Carroll, 1979; Cavaco & Crifo, 2014). Second, given that firm-level CSR scores were not available for the majority of companies in our sample, most of which are small and mid-sized French firms not individually rated, the EcoVadis sectoral benchmarks provided the most reliable and internally consistent proxy for the CSR environment in which each firm operates. Third, the annual updating of EcoVadis sectoral averages allows for temporal variation in the CSR score, capturing the progressive diffusion of non-financial standards across industries over the 2021–2024 period. While this approach does not capture intra-sector heterogeneity at the firm level, it reflects the industry-level CSR maturity that frames each firm's non-financial operating environment, a methodological choice that is both transparent and replicable.

The collection of CSR data relied on a two-stage pipeline combining automatic text classification and sectoral score attribution. This pipeline consists of four main stages: keyword-based classification, text normalization, score mapping, and integration.

[5] EcoVadis. *EcoVadis Sustainability Ratings*. Retrieved from https://ecovadis.com/

1. Data extraction and classification: Classification was performed using a Python script applying a text mining procedure to the APE label (*libellé APE*) of each firm. A dictionary of CSR keywords, structured around the EcoVadis framework, was used to match each label against four main CSR components: **Environment**, **Labor & Human Rights**, **Ethics**, and **Sustainable Procurement**. Each component was further divided into sub-themes ; for instance, *Energy consumption & GHG emissions*, *Water*, *Biodiversity*, or *Pollution* under the Environment pillar. The classifier assigned to each firm the (component, sub-theme) pair with the highest number of keyword matches.

2. Normalization and Standardization. The APE labels were originally written in various formats, with differences in accents, apostrophes, and capitalization. To ensure consistent and reliable keyword matching, we applied a systematic normalization procedure. The `normalizer`[6] function transformed each label into a standardized string through the following steps: converting to lowercase, decomposing Unicode characters and removing diacritics, removing apostrophes, and collapsing multiple spaces into a single space.

3. Feature engineering: Based on the classified CSR component and sub-theme, each firm was mapped to an EcoVadis sector and theme pair, enabling the construction of a quantitative CSR score variable. These include:

- **Annual CSR scores** (`Score_CSR_2021` to `Score_CSR_2024`), corresponding to the EcoVadis sectoral average for the relevant (theme, sector, year) combination;
- **Four-year average CSR score** (`Score_CSR_Moyen_4ans`), computed as the arithmetic mean of available annual scores;
- **EcoVadis sector and theme labels** (`Secteur_EcoVadis`, `Theme_EcoVadis`), capturing the sectoral CSR context of each firm.

4. Integration and merging of CSR variables: The newly constructed CSR variables were merged with the initial firm-level database using the **SIREN** number as

[6] The `normalizer` function is a custom Python function applying four sequential transformations: lowercasing, Unicode diacritic removal (`unicodedata.normalize("NFD")`), apostrophe and quotation mark deletion, and whitespace normalization. For instance, `"Fabrication d'équipements électriques"` becomes `"fabrication dequipements electriques"`.

the merging key, ensuring reliable alignment between sources.

Finally, the CSR and financial datasets were merged into a unified panel database structured in long format, where each observation corresponds to a unique (**SIREN**, **year**) pair. The merging key combining firm identifier and fiscal year ensured reliable alignment between the two sources across the 2021–2024 period.

Three additional normalized indicators were computed at this stage to facilitate econometric modelling:

- a **net profitability ratio**, defined as net income scaled by total assets (*rentabilité nette*);
- a **debt ratio**, defined as financial debt scaled by total assets (*taux d'endettement*);
- a **log-transformed total assets** variable, introduced as a firm size control.

The resulting consolidated dataset comprises **2,997 observations** across **1,005 firms** and **26 variables**.

3.2 Descriptive Statistics

3.2.1 Financial data

Revenue

The distribution of revenue is highly skewed. The mean revenue reaches € 84.4 million, while the median is only € 5.06 million. This large gap indicates the presence of a few very large firms that pull the average upward, while most companies operate at much smaller scales. The extreme values — ranging from –€ 45,600 to € 16.9 billion — confirm the strong heterogeneity of firm sizes.

Net Income

A similar pattern is observed for net income. The mean is € 4.95 million, but the median is only € 590,000, showing that most firms generate modest profits while a minority report exceptionally high earnings (up to € 3.67 billion). The presence of large losses (minimum –€ 409 million) also contributes to the high standard deviation.

Financial Debt

Financial debt follows the same logic: an average of € 66.1 million contrasts with a median of € 674,000, indicating that most firms carry limited debt while a few exhibit extremely high leverage (up to € 14.8 billion).

Profitability Ratios

The profitability ratios show substantial dispersion. The average ROE is strongly negative (–736.653%), with a median of 9.2%. This contrast reveals the presence of extreme negative values (down to –1,370,000%), typically caused by firms with very low or negative equity. Similarly, ROA has a mean of –590.554% and a median of 2.2%, reflecting the influence of extreme outliers.

Total Assets

Total assets also exhibit a highly skewed distribution: the mean is € 315.8 million while the median is € 5 million. The maximum value reaches € 71 billion, confirming the coexistence of very small and extremely large firms within the sample.

Net Profitability Ratio

The net profitability ratio shows a mean of 1.472 and a median close to zero (0.024), indicating that most firms operate with very low profitability, while a few achieve exceptionally high returns (up to 3,814%).

Debt Ratio

The debt ratio presents a similar pattern: a mean of 11.282 but a median of only 0.224, again highlighting the presence of extreme values (up to 25,285%).

Log-Transformed Total Assets

The variable `log_total_actif` corresponds to the logarithmic transformation of total assets. It shows a mean of 15.599 and a median of 15.425. This transformation helps scale total assets relative to firm size, reduces the asymmetry caused by extreme capitalization differences, and produces a more regular distribution. It also facilitates model interpretation by limiting the disproportionate influence of very large firms.

Variable	**Mean**	**Std. Dev.**	**Min**	**Max**	**Median**
Identifiers					
CSR Score	57.837	4.600	45.5	66.6	58.3
Financial figures (absolute values, €)					
Revenue	84,400,328	83,471,754	−45,600	1,690,000,000	50,600,000
Net Income	4,953,023	96,308,103	−40,900,000	367,000,000	59,000
Financial Debt	66,110,078	688,511,361	0	14,800,000,000	674,000
Total Assets	315,800,028	3,371,138,274	426	71,000,000,000	5,000,000
Ratios & transformed variables					
ROE (%)	−736.653	31,896.933	−137,000	8,140	9.2
ROA (%)	−590.554	25,377.751	−109,000	10,100	2.2
Net Profit Margin	1.472	76.170	−25.368	3,814.286	0.024
Debt Ratio	11.282	525.754	0	25,285.714	0.224
Log Total Assets	15.599	2.338	6.057	24.986	15.425

Table 1: Descriptive statistics of financial and CSR variables.

3.2.2 CSR Score Analysis

As noted in the data construction section, the CSR score is not a firm-level measure but a sector $\times$ year benchmark derived from the EcoVadis rating framework. Each firm is assigned the average CSR score of its corresponding EcoVadis sector and thematic category for a given year. This construction reflects the idea that CSR expectations and regulatory pressures are largely sector-specific, and that firms operate within a shared CSR environment shaped by industry-level norms.

This construction carries two methodological implications. First, the CSR score should be interpreted as a contextual CSR indicator capturing the non-financial environment in which firms operate, rather than their individual CSR performance. It reflects sectoral maturity in CSR practices, exposure to sustainability risks, and the diffusion of responsible business standards within each industry. Second, because the score varies only across sectors and years, it absorbs structural differences in CSR intensity that are common to firms within the same industry. This reduces measurement noise and mitigates firm-level reporting biases, but it also limits the ability to capture intra-sector heterogeneity.

The CSR score should therefore be understood as a sector-adjusted proxy for the CSR context surrounding each firm, rather than a direct measure of firm-level CSR engagement. This also provides a structural explanation for the near-zero correlations observed between the CSR score and financial variables as will be shown in Figure 3 below: since the score varies only across sectors and years, its within-sector variance is by construction close to zero, mechanically compressing its covariance with firm-level financial outcomes.

Distribution and Temporal Trends

The CSR score presents a concentrated distribution (45.5–66.6 points), with a mean of 57.84 and a median of 58.30, reflecting a slight positive skew. The standard deviation of 4.60 points confirms the relative homogeneity of the sample. Over the 2021–2024 period, a moderate but continuous improvement in scores is observed, consistent with the growing maturity of CSR practices.

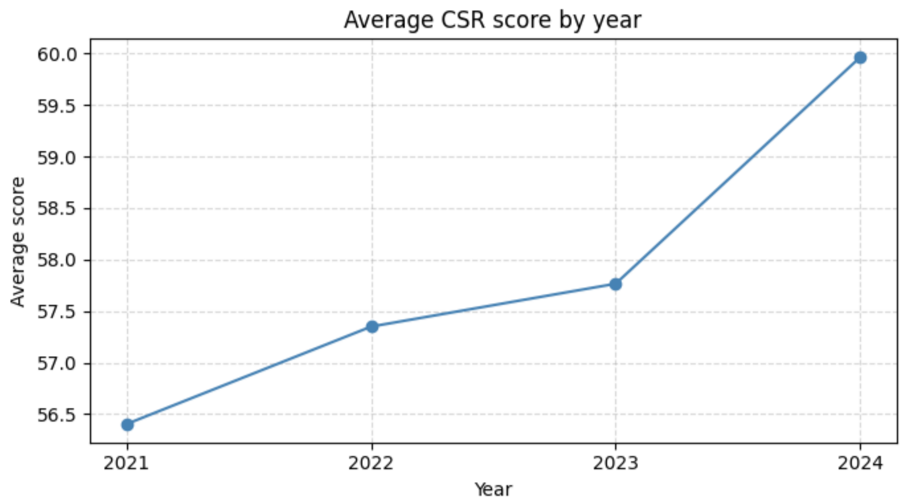

Figure 1: Evolution of the Average CSR Score from 2021 to 2024

Sectoral Disparities

Sector-level analysis (EcoVadis sectors with ≥ 10 observations) reveals significant differences between sectors. The 95% confidence intervals allow the identification of sectors that statistically deviate from the overall mean. These disparities justify the inclusion of sector-level controls in subsequent multivariate analyses.

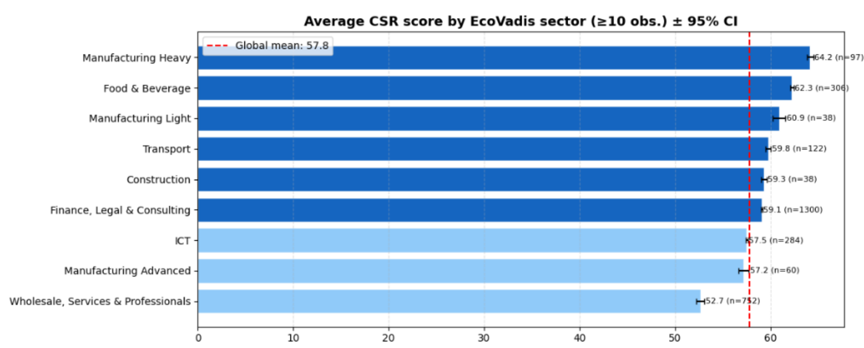

Figure 2: Sectoral Disparities in CSR Performance

Correlation Structure

Analysis of the correlation matrix alone reveals the absence of significant linear correlations between CSR scores and financial variables. The low or near-zero correlations between the CSR score and all financial dimensions suggest that extra-financial performance cannot be reduced to economic performance alone, and that CSR engagement follows a logic largely independent of firms financial characteristics.

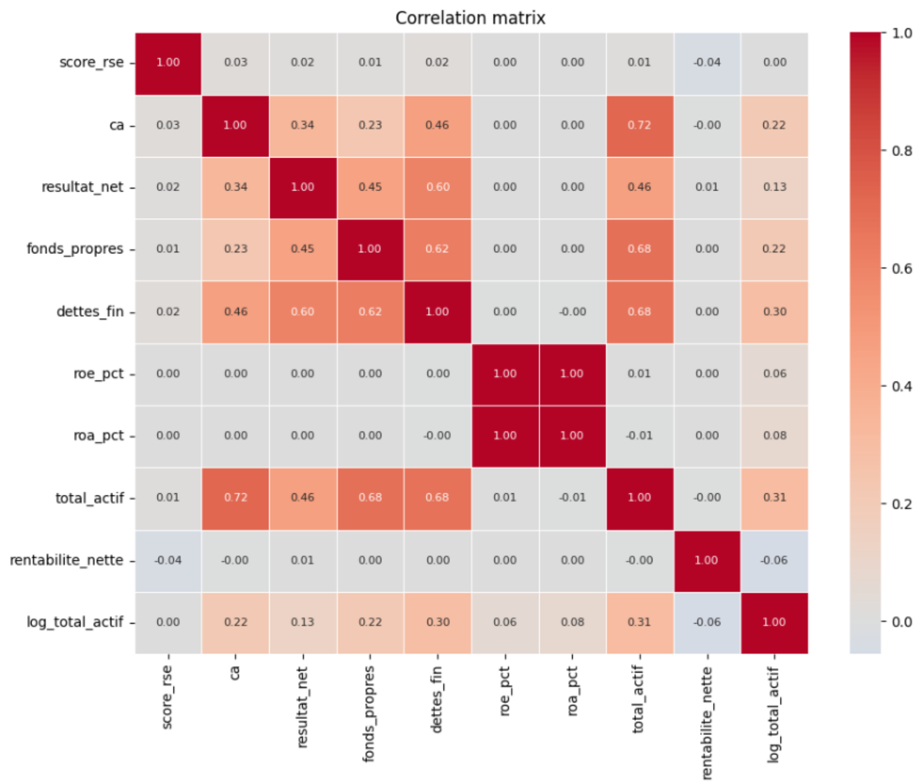

Figure 3: Correlation Structure of CSR and Financial Dimensions

Based on the correlation matrix, the *score_rse* appears uncorrelated with all financial variables, with coefficients ranging from -0.04 to 0.03, suggesting that, at least linearly, CSR performance does not seem to be driven by firms' financial characteristics.

3.2.3 Data Quality

Two categories of variables can be distinguished according to their completion rate. Highly incomplete variables ($> 30\%$ missing) include primarily *ca_export* (46.4%), *ca* (40.4%), and the profitability ratios (ROA, ROE, EBIT [7] $\approx 38\%$), due to the absence of mandatory reporting requirements for certain legal entities. In contrast, the core vari-

[7] EBIT : Earnings Before Interest and Taxes

ables of the analysis -*score_rse*, *annee*, *SIREN*, and *Secteur_EcoVadis* are fully complete (0% missing).

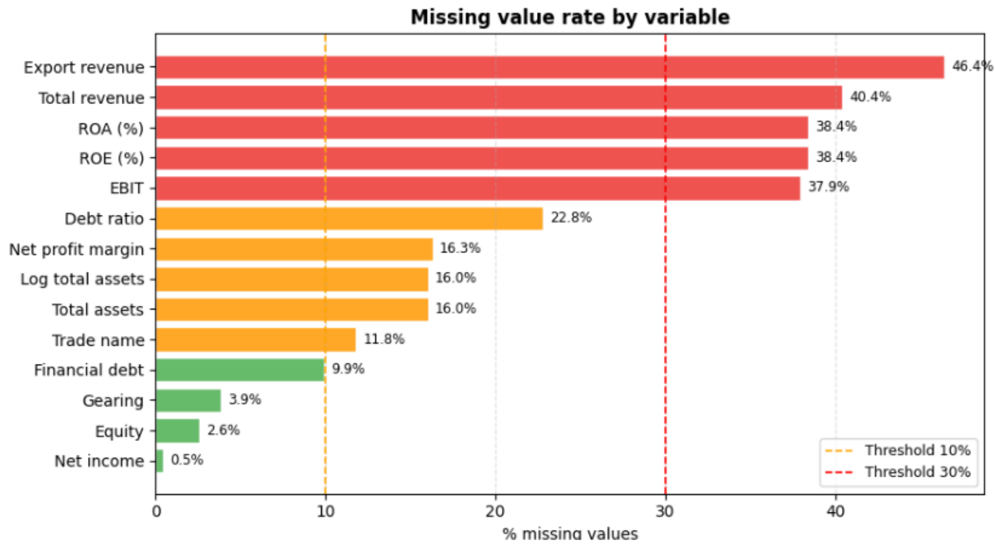

Figure 4: Distribution of Missing Data Across Variables

The descriptive analysis highlights several structural characteristics of the dataset. Financial variables such as revenue, net income, financial debt, and total assets display strongly right-skewed distributions, with large gaps between means and medians. This indicates the coexistence of many small firms alongside a limited number of very large corporations with extreme values. These properties justify the use of robust statistical methods and log-transformations to limit the influence of outliers in subsequent analyses. CSR scores, by contrast, show a more homogeneous distribution and a gradual increase over the 2021–2024 period, although noticeable sectoral differences suggest that industry-specific factors influence CSR engagement.

While some financial variables contain substantial missing values, particularly export revenue and certain profitability indicators, the main analytical variables remain complete, preserving the reliability of the panel dataset. Overall, these descriptive results provide a solid basis for the empirical analyses that follow.

4 Results and Discussion

4.1 Model Comparison

4.1.1 Two-Way Fixed Effects (TWFE) Baseline Estimates

To estimate the effects of Mission-Driven Firms certification on firm financial performance, two complementary difference-in-differences estimators were implemented: the classical Two-Way Fixed Effects (TWFE) model, and the Sun & Abraham (2021) estimator, which will constitute the preferred specification of this study.

The TWFE model regresses each financial outcome on a binary post-treatment indicator, defined as the acquisition of the *Mission-Driven Firms* status, controlling for firm-level fixed effects, year fixed effects, and a set of macroeconomic controls (GDP growth, inflation, ECB refinancing rate, and industrial production index). Results are presented across six outcomes (net income, revenue, EBIT, ROE, economic profitability, and gearing) with standard errors clustered at the firm level to account for within-firm serial correlation.

$$Y_{it} = \beta \cdot \text{Post}_{it} + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it}$$

All TWFE estimates are statistically insignificant at conventional levels ($p > 0.10$) for five of the six outcomes considered, with p-values ranging from 0.104 for Net Income to 0.960 for EBIT. The only exception is economic profitability, for which the estimated effect is negative and statistically significant ($\beta = -0.054$, $p = 0.004$). This result should nevertheless be interpreted with caution given the number of outcomes tested and the known limitations of the TWFE estimator in staggered-adoption settings. Overall, the TWFE estimates provide little evidence that the acquisition of *Mission-Driven Firm* status has a significant effect on firms' financial performance in the short-term period following treatment. As noted above, TWFE is reported here primarily as a benchmark estimator.

Variable	β (post)	SE	p-val	Sig.	N
Net income (M€)	−0.8935	0.5493	0.1040	n.s.	2,065
Revenue (M€)	+6.6011	6.2715	0.2929	n.s.	1,253
EBIT (M€)	−0.0267	0.5262	0.9596	n.s.	1,305
ROE (%)	+3.7085	11.0817	0.7380	n.s.	1,296
Economic profitability (%)	−0.0542	0.0189	0.0042	***	1,787
Gearing	−0.1436	0.3296	0.6631	n.s.	2,011

Table 2: TWFE Estimates

*** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$ n.s. = not significant

4.1.2 Event Study and Pre-Trend Validation

The event-study extension of the TWFE confirms this pattern. Due to the limited time span of the panel (2021–2024), the dummy at $t-3$ is absorbed by the year fixed effects and cannot be estimated. Pre-treatment coefficients are therefore only available at $t-2$, and are uniformly non-significant across all three outcomes (net income, ROE, gearing), supporting the parallel trends assumption visually. This is formally corroborated by F-tests on the sole available pre-treatment coefficient: $F(1,1442) = 1.538$ ($p = 0.215$) for net income, $F(1,868) = 0.257$ ($p = 0.612$) for ROE, and $F(1,1396) = 0.857$ ($p = 0.355$) for gearing. In all three cases, the null hypothesis of parallel trends prior to treatment cannot be rejected. The estimated coefficients for the three dependent variables remain statistically insignificant across all post-treatment periods and specifications.

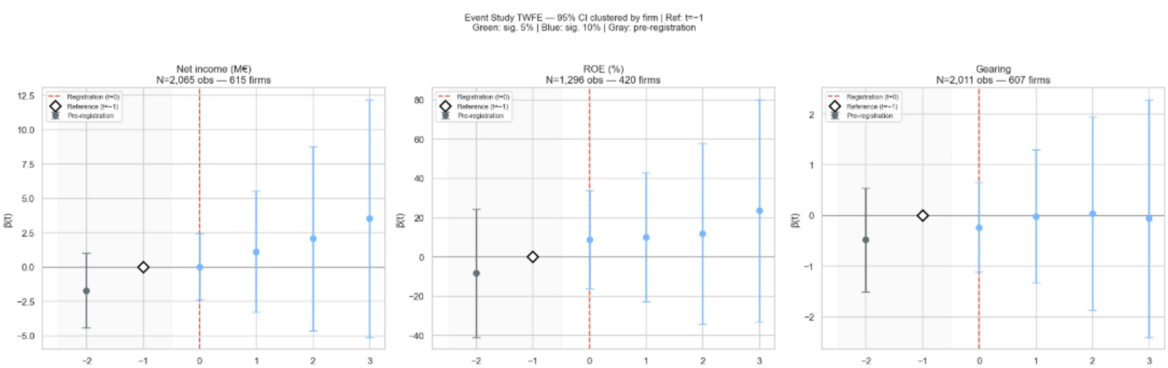

Figure 5: Event Study TWFE

4.1.3 Sun & Abraham Estimator and Cohort Heterogeneity

The Sun and Abraham (2021) estimator was then used as the primary specification. Comparison of the aggregate Average Treatment Effect on the Treated (ATT) estimates reveals divergences from the TWFE estimates of varying magnitude across the three indicators. For net income, the S&A estimate (−1.024M) is modestly larger in absolute value than the TWFE estimate (−0.894M), a difference of approximately 15%. For ROE, both estimators yield negative estimates, but of very different magnitudes (TWFE: +3.709 percentage points; S&A: −1.800 percentage points), with a sign reversal. For gearing, the two estimates are close in sign and magnitude (TWFE: −0.144; S&A: −0.039), though the S&A estimate is substantially smaller. The sign reversal on ROE and the magnitude divergence on net income both exceed the 20% threshold generally used to signal significant heterogeneity between cohorts[8], confirming that the TWFE estimate is likely biased in this context of staggered adoption and that the Sun and Abraham estimator should be preferred.

Variable	**ATT (S&A)**	**ATT (TWFE)**	**Δ** (S&A - TWFE)	**Interpretation**
Net Income (M€)	−1.0237	−0.8935	−0.1302	△ Cohort heterogeneity
ROE (%)	−1.7997	+3.7085	−5.5082	△ Cohort heterogeneity (sign reversal)
Gearing	−0.0385	−0.1436	+0.1051	Limited heterogeneity

Table 3: Aggregated ATT(g, t) (weighted by cohort × period number of observations)

Robustness checks across four alternative specifications : parsimonious control set, alternative winsorization[9] bounds (P5–P95), a narrower ±2-year event window, and a stricter minimum post-treatment observation requirement. It's consistently yield non-significant estimates for net income at conventional levels, with coefficients ranging from −0.178 to −0.914. The one exception is the log(net income) specification, which is significant at the 10% level ($p = 0.078$) but applies only to the subsample of profitable firms and is not directly comparable to level estimates. The stability of results across all variants strengthens confidence that no measurable average treatment effect is present in the full sample.

[8]Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277. The paper shows that two-way fixed effects estimators can be biased when treatment timing is staggered due to heterogeneity across treatment cohorts.

[9]Winsorization is a statistical technique that limits the influence of extreme values by capping observations below a lower percentile and above an upper percentile threshold, rather than removing them entirely.

Specification	$\boldsymbol{\beta}$(post)	p-value	Sig.	N obs	N firms
Main spec. (W P1–P99)	−0.8935	0.1040	n.s.	2,065	615
Parsimonious controls	−0.8935	0.1040	n.s.	2,065	615
Winsorisation P5–P95	−0.1783	0.3480	n.s.	2,065	615
Narrow window (±2 years)	−0.9144	0.1373	n.s.	1,748	612
Min. ≥3 post-treatment obs.	−0.6448	0.4680	n.s.	1,259	346
log(Net income) – △ profitable subsample	−0.2757	0.0776	*	1,301	482

Table 4: Robustness Checks – Net Income

*** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$ n.s. = non-significant

4.2 Interpretation of the Sun & Abraham DiD Model

4.2.1 Estimator mechanics

The standard TWFE estimator computes a single treatment effect $\hat{\beta}_{\text{TWFE}}$ by implicitly aggregating all 2×2 comparisons across cohorts over time. As shown by Sun and Abraham (2021), in a staggered adoption setting, this coefficient can be expressed as a weighted sum of cohort-specific effects:

$$\hat{\beta}_{\text{TWFE}} = \sum_{g,\ell} \omega^{\ell}_{g,\ell'} \cdot \text{CATT}_{g,\ell}$$

where the weights $\omega_{g,\ell}$ are nonlinear functions of cohort sizes and timing. These weights may be negative and may place weight on periods $\ell' \neq \ell$, implying that the estimated effect for a given period can be contaminated by effects from other periods. This issue invalidates the causal interpretation of TWFE estimates when treatment effects are heterogeneous across cohorts.

The Sun & Abraham estimator addresses this limitation by estimating each cohort-specific average treatment effect:

$$\text{CATT}_{g,\ell} = \mathbb{E}\left[Y_{i,g+\ell} - Y^{\infty}_{i,g+\ell} \mid E_i = g\right]$$

where $Y^{\infty}_{i,t}$ denotes the potential outcome in the absence of treatment. This parameter captures the average difference between observed outcomes and the counterfactual outcome for cohort g at relative time ℓ.

This estimator can be interpreted as a two-period difference-in-differences:

$$\hat{\delta}_{g,\ell} = \left(\bar{Y}_{g,g+\ell} - \bar{Y}_{g,g-1}\right) - \left(\bar{Y}_{C,g+\ell} - \bar{Y}_{C,g-1}\right)$$

where $\bar{Y}_{g,t}$ is the average outcome for cohort g at time t, $g-1$ is the pre-treatment reference period, and C denotes the control group composed of not-yet-treated units.

4.2.2 Aggregation and weighting

The cohort-time specific effects are then aggregated into an overall ATT:

$$\hat{\nu} = \sum_{\ell \geq 0} \sum_{g} \hat{\delta}_{g,\ell} \cdot \hat{w}_{g,\ell}, \qquad \hat{w}_{g,\ell} = \frac{N_{g,\ell}}{\sum_{g'} N_{g',\ell}}$$

where the weights $w_{g,\ell}$ are proportional to the number of observations $N_{g,\ell}$ in each (g, ℓ) cell. This convex weighting scheme ensures that the aggregate estimate lies within the support of the underlying cohort effects and prevents small cohorts from being overweighted. Cells that are not identified are excluded from the aggregation.

4.2.3 Aggregate results

The aggregated ATT estimates are as follows: -1.024 million euros for net income, -1.800 percentage points for ROE, and -0.039 for gearing. While the net income and ROE estimates are directionally consistent with a modest negative short-term effect following certification, none of these effects are statistically significant at conventional levels.

This lack of statistical significance reflects the low statistical power of the dataset, which includes 616 firms with an average of 3.4 observations per firm. As a result, the estimated cohort-specific effects $\hat{\delta}_{g,\ell}$ are imprecisely estimated, leading to wide confidence intervals.

4.2.4 Directional heterogeneity across cohorts

The divergence $|\hat{\nu}_{S\&A} - \hat{\beta}_{\text{TWFE}}| > 20\%$ of $|\hat{\beta}_{\text{TWFE}}|$ for ROE, where the two estimators yield estimates of opposite sign ($+3.709$ versus -1.8 percentage points), signals sufficiently strong heterogeneity in effects across cohorts for the TWFE estimator to be potentially biased for this outcome. For net income and gearing, the divergence between the two estimators is more limited, though the sign reversal on ROE alone is sufficient to motivate the use of the S&A estimator as the preferred specification. The ATT(g, t) plots by cohort confirm this interpretation: no cohort generates a clear, directional, and persistent post-treatment effect; cohort-specific trajectories differ substantially in both sign and magnitude depending on the time window considered. Firms that adopted early (cohorts 2019–2021) exhibit structurally distinct profiles, in size, sector, and pre-treatment financial trajectory, from those that adopted later (cohorts 2022–2023), making the assumption of homogeneous effects particularly implausible. The S&A estimator is specifically designed to make this heterogeneity visible rather than masking it within an aggregated $\hat{\beta}$.

4.2.5 Identification validity and residual selection bias

A potential concern for identification arises from the non-zero weighted pre-treatment mean for net income (+1.805 million). In the Sun & Abraham framework, this statistic is computed by comparing each cohort to not-yet-treated cohorts at the same date, making it more sensitive to heterogeneous pre-trends than TWFE.

This result suggests that firms acquiring the *Mission-Driven Firm* status early and those acquiring it later may not have followed similar financial trajectories prior to treatment, which raises concerns about the validity of the conditional parallel trends assumption required for causal interpretation.

Moreover, in the absence of a never-treated group, this assumption cannot be directly tested. Late adopters serve as controls for early adopters only if their counterfactual trends would have evolved similarly in the absence of treatment, an assumption that is difficult to justify when treatment timing may be endogenous to firm performance.

Overall, the Sun & Abraham estimates suggest the absence of a statistically significant average effect, combined with substantial heterogeneity across cohorts. Wide confidence intervals reflect both limited statistical power and variability in treatment effects, preventing strong causal conclusions.

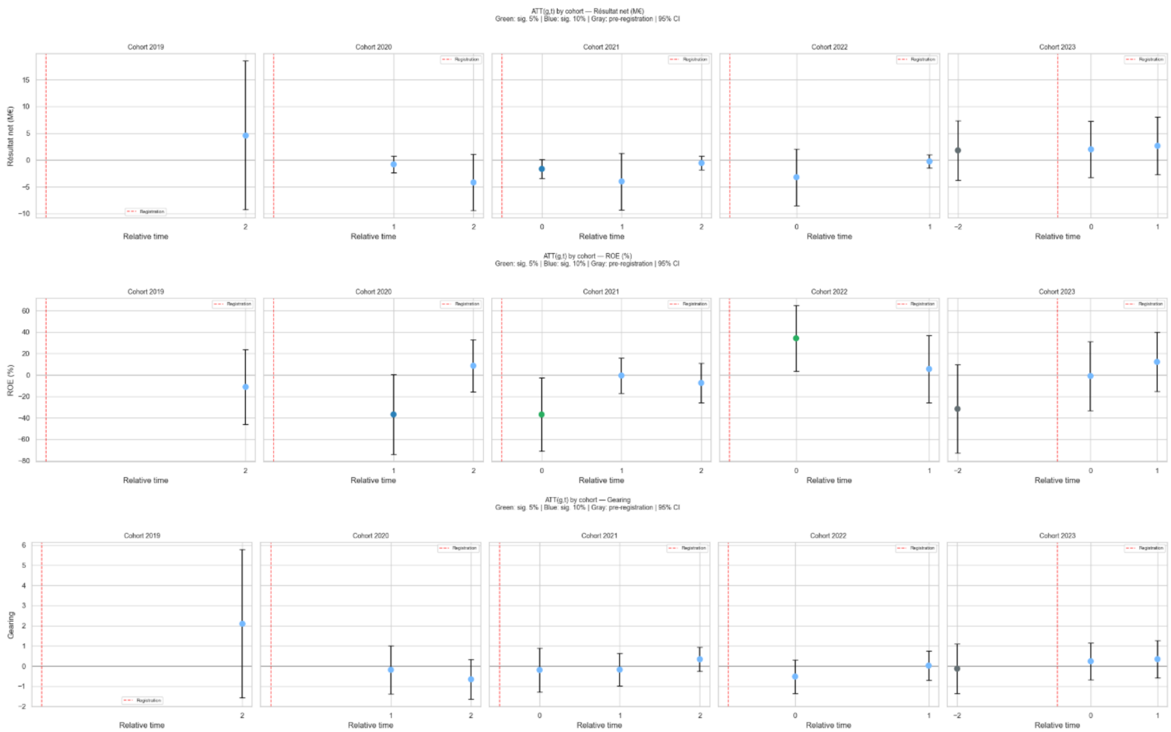

Figure 6: ATT by cohorts & by variables

4.3 Reflections and Limitations

Several structural limitations of this study must be acknowledged in order to properly bound the scope and interpretation of the findings.

4.3.1 Absence of a never-treated control group

The primary identification challenge stems from the absence of a never-treated control group. The sample consists exclusively of firms that eventually adopt the *Mission-Driven Firms* certification. As a result, identification relies entirely on variation in treatment timing: early adopters serve as controls for later adopters prior to their certification.

This staggered adoption framework imposes a stronger assumption than the standard parallel trends assumption. In particular, it requires that, in the absence of treatment, the outcome trajectories of early and late adopters would have evolved similarly over time. This assumption is not directly testable.

If firms time their certification strategically, such as adopting after strong performance or in response to anticipated downturns, the counterfactual constructed from not-yet-treated firms may be invalid. The non-zero pre-treatment mean observed in the Sun & Abraham framework for net income is consistent with this concern and suggests potential selection bias.

4.3.2 Limited panel depth and statistical power

The dataset includes 616 firms observed over the 2019–2024 period, with an average of only 3.4 observations per firm. This limited panel depth constrains the ability to precisely estimate treatment effects.

In addition, several cohort-time cells $\mathrm{ATT}(g, t)$ could not be identified due to insufficient observations, leading to the exclusion of 14 cells from the estimation. This further reduces the effective degrees of freedom.

As a consequence, confidence intervals are wide across all specifications, making it difficult to distinguish between a true absence of effect and an economically meaningful effect that the data lacks the power to detect. Expanding the time horizon and incorporating a larger sample, ideally including comparable non-treated firms would substantially improve statistical precision.

4.3.3 Measurement and selection of financial outcomes

Winsorization at the P1–P99 level reduces the influence of extreme values, but several variables, particularly net income and EBIT, remain highly skewed.

Logarithmic transformations model produce statistically significant estimates for net income; however, they are only applicable to observations with strictly positive values, which represent approximately 63% of the sample. This restriction introduces a selection bias toward financially healthier firms. Consequently, log(net income) is the only variable for which the treatment effect reaches statistical significance in the robustness checks. The economic profitability also yields a significant estimate in the baseline TWFE specification ($\beta = -0.054$, $p = 0.004$). Both results must be interpreted with caution: the log specification applies only to a selected subsample of profitable firms, while the economic profitability estimate, though statistically significant, represents a negligible effect size.

This marginal significance can be understood in two complementary ways. First, net income is a relatively responsive indicator that captures short-term operational adjustments following certification, such as improved stakeholder relations, reputational gains, or enhanced employee engagement. These mechanisms are consistent with stakeholder theory and with empirical findings showing that CSR can generate incremental performance gains through trust-based interactions. Second, the logarithmic transformation itself reduces the influence of extreme values, which are particularly prevalent in this dataset, meaning that the significant effect reflects a broad distributional shift in profitability rather than the influence of a few outlying firms.

As a result, estimates obtained from level and log specifications are not directly comparable. Rather than constituting robustness checks, they reflect treatment effects on structurally different subsamples. Log-based results should therefore be interpreted as indicative of effects among financially stronger firms, rather than as evidence of a general effect across the full sample.

4.3.4 Data reliability and treatment of outliers

A complementary limitation concerns the reliability of the raw financial data and the presence of atypical values within the initial dataset. A systematic cross-verification conducted directly on the data source (*Pappers*) and on *Verif* confirmed that these inconsistencies originate from the initial filing of tax statements by registries or firms themselves, rather than from any algorithmic bias introduced during the scraping process.

Regarding performance indicators, particular attention was paid to measurement units to ensure analytical consistency. Profitability ratios, namely ROE (Return on Equity) and ROA (Return on Assets), were adjusted to be expressed in percentage terms (for example, a value of 12 corresponding to 12% rather than 0.12). This methodological choice, implemented through a multiplication by 100 in the Python processing script, allows for a more direct interpretation of the results and avoids common misreadings associated with decimal notation. This standardization strengthens the robustness of the statistical comparisons conducted in the remainder of the study.

4.3.5 Sectoral heterogeneity

The sample spans a wide range of industries, raising the possibility that the effect of certification varies across sectors.

Sector-specific estimations reveal no statistically significant effect on net income across most of the main sectors represented in the data. The sole exception is Finance, Legal & Consulting, where the estimate is significant at the 10% level ($\beta = -1.849$, $p = 0.061$). Estimated coefficients range from -1.849 to $+2.033$, with p-values above conventional thresholds for all remaining sectors.

Sector	$\boldsymbol{\beta}$**(post)**	**p-value**	**Sig.**	**N obs**	**N firms**
Finance, Legal & Consulting	−1.8490	0.0606	*	877	260
Wholesale, Services & Professionals	−1.6120	0.1077	n.s.	539	161
Food & Beverage	+2.0333	0.2966	n.s.	215	65
ICT	+0.4467	0.2734	n.s.	205	61
Transport	+0.6400	0.8296	n.s.	87	26

Table 5: Heterogeneity by Sector (EcoVadis) – Net Income

*** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$ n.s. = non-significant

While only one sector yields a marginally significant estimate, these results do not provide robust evidence of broad sector-specific effects, and their interpretation is further limited by reduced sample sizes within each sector, which weakens statistical power.

4.3.6 External validity

The external validity of the findings is limited. The results apply specifically to firms that chose to adopt the Mission-Driven Firms certification between 2019 and 2024.

The sample is not representative of the broader population of firms, either in France or internationally, nor of ESG certification schemes in general. Institutional, regulatory, and strategic differences across contexts may lead to different effects.

Caution is therefore required when attempting to generalize these findings beyond the specific setting studied.

Overall, while the empirical strategy is methodologically rigorous relying on the Sun &

Abraham estimator, formal tests of identifying assumptions, and multiple robustness checks, the structural limitations of the data prevent definitive causal conclusions.

The results are best interpreted as providing no strong evidence of a detectable average financial effect of *Mission-Driven Firms* certification in the short to medium run, while leaving open the possibility of heterogeneous, cohort-specific, or longer-term effects that the current data is not able to capture.

5 Conclusion

This study examines the relationship between Corporate Social Responsibility (CSR) and Corporate Financial Performance (CFP) through the lens of *Mission-Driven Firm* certification — a legal status introduced by the French PACTE Law of 2019 that formally embeds social and environmental commitments into a firm's corporate purpose. The central question animating this research is whether this institutionalized form of CSR commitment translates into measurable financial outcomes. Drawing on Stakeholder Theory, which posits that CSR engagement tends to enhance CFP albeit not unconditionally, and building on a substantial body of prior empirical work, this research investigates the extent to which the formal adoption of *Mission-Driven Firm* status contributes to improved firm-level financial performance.

The empirical analysis relies on a firm-level panel dataset combining financial performance indicators with an industry-adjusted CSR measure constructed via text mining. A difference-in-differences framework was employed to account for the staggered adoption of certification across firms in the sample, thereby controlling for pre-existing heterogeneity between treated and untreated entities. Baseline estimates are complemented by a series of robustness checks and an alternative specification designed to address cohort heterogeneity.

The results yield no statistically significant evidence of an average financial impact associated with Mission-Driven Firm certification across five of the six performance metrics considered (Net Income, Revenue, EBIT, ROE, and Gearing). The sole exception is economic profitability, which yields a statistically significant but economically negligible estimate in the baseline TWFE specification. Consistent with this finding, neither event study analyses nor robustness checks reveal average treatment effects of meaningful magnitude at the full-sample level, over either the short or medium term. That said, the evidence does point to substantial heterogeneity in the financial consequences of CSR adoption across cohorts, suggesting that the CSR–CFP relationship is far from uniform across firms.

Several limitations bear on the interpretation of these findings: the absence of never-treated control firms, a relatively short panel, and data quality constraints that remain difficult to fully resolve. The use of text mining, while practical, does not eliminate the inherent challenges of CSR measurement. External validity is further bounded by the sample's restriction to French firms that obtained certification between 2019 and 2024.

Importantly, the absence of detectable short-term financial effects does not preclude other value creating mechanisms through which CSR may operate notably reputational capital, enhanced stakeholder trust, or long-term strategic positioning. Preliminary findings also raise the possibility that firms with stronger ex ante financial profiles respond differently to certification, pointing to temporal dynamics and heterogeneity effects that warrant further investigation.

Future research would benefit from longer observation windows, more credible counterfactuals incorporating never-treated firms, and more granular measures of CSR engagement. Such extensions would help clarify whether certification effects materialize with a delay, or whether CSR certification operates principally through indirect, non-financial pathways.

Taken together, while no short-term financial premium from *Mission-Driven Firm* certification is apparent in this data, the study makes a meaningful contribution to the empirical CSR literature by generating causal estimates using econometric methods suited to staggered treatment designs. More broadly, these results highlight both the complexity of identifying the financial returns of CSR and the crucial importance of accounting for identification challenges and firm-level heterogeneity. In the specific context of Mission-Driven Firms, where CSR is not merely a reputational signal but a statutory commitment embedded in corporate governance, this complexity is particularly salient: the absence of a detectable short-term effect does not diminish the strategic or institutional significance of the *Mission-Driven Firm* status, but it does underscore that financial benefits, if they exist, are likely to materialize over longer horizons and through indirect pathways that remain to be identified.

Annex

Table 6: Sample Selection — Successive Restrictions

Step	**N obs.**	**N firms**
DATASET CONSTRUCTION		
Initial dataset (after scraping)	2,997	1,005
Exclusion of 2024–2025 cohorts	2,308	778
Filter: at least 2 post-treatment observations per firm	2,073	616
Final analytical panel (winsorisation P1–P99)	2,073	616
ESTIMATION SUBSAMPLES		
TWFE / S&A — Net income (M)	2,065	615
TWFE / S&A — Revenue (M)	1,253	413
TWFE / S&A — EBIT (M)	1,305	420
TWFE / S&A — ROE (%)	1,296	420
TWFE / S&A — Economic return (%)	1,787	552
TWFE / S&A — Gearing	2,011	607
RESTRICTED SUBSAMPLE		
log(Net income) — strictly positive values only	1,301	482

Note: The number of observations used in TWFE/S&A estimations may differ slightly from the raw sample counts because of internal filtering procedures in `PanelOLS` and `feols`, including the absorption of empty fixed-effect cells and the application of the $\pm$3-year event window.

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